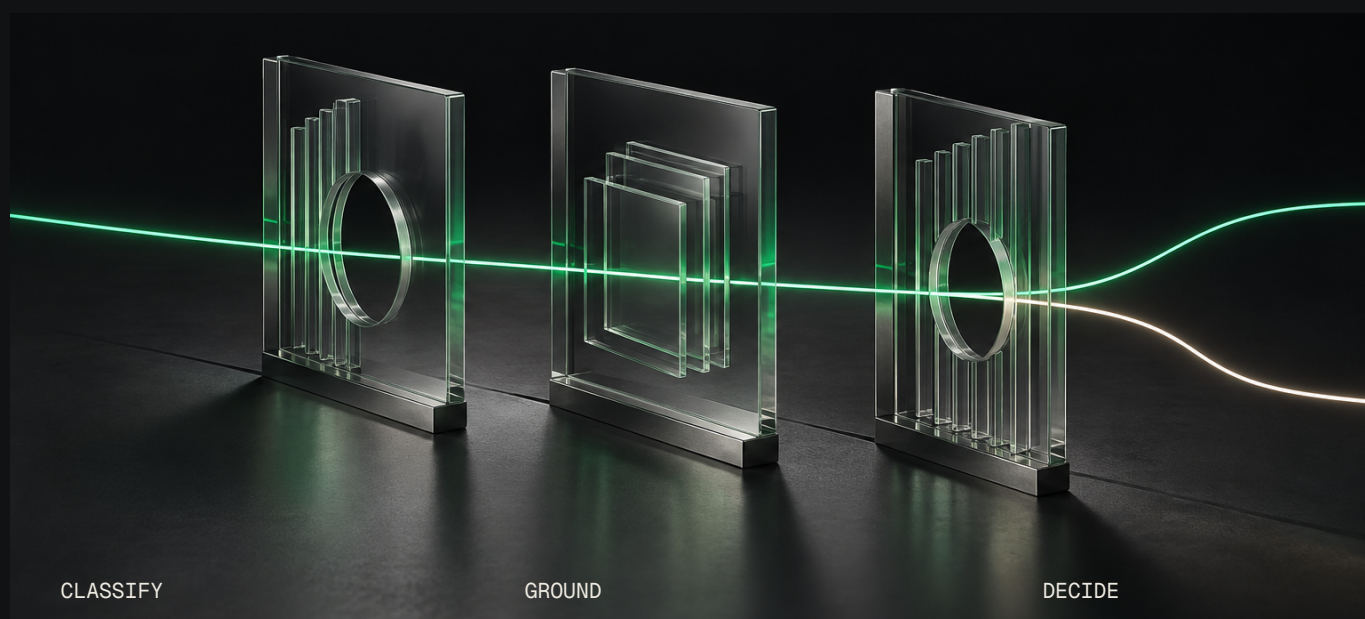


HIVER SDE INTERN TAKE-HOME / 17 SEPTEMBER 2026

An AI support agent that earns its answer.

Built and reviewed by Arnav Bule. Cadence classifies customer messages, drafts grounded replies and explains when a human must take over.



27,627

Real support threads

SpotifyCares conversations from the 2017 Twitter support corpus.

200

Frozen benchmark cases

Four matched systems. Labels and scores reviewed by Arnav Bule.

60

New confirmation cases

Separated from inspected data; AI labels frozen before inference.

The engineering question: How much support can the system usefully handle automatically while missing as few necessary escalations as possible? Fluency and intent accuracy alone do not answer that question.

[Live demo](#)[Repository and runnable evidence](#)

Start with the customer.

A small, inspectable pipeline; explicit labels; separate evidence for each version.

PROBLEM FRAMING

SpotifyCares has recurring technical questions and account-specific requests. “Good” means relevant, supported, useful and appropriately escalated. The agent drafts public replies; it has no account tools and never posts tweets, refunds money or contacts staff.

Twelve intents cover playback, playlists, plans, billing, login, security, downloads, catalog, artist metadata, feedback, language and other requests. Billing/security defaults and deterministic safety rules override model confidence. The threshold is frozen at 0.90.

TWO IMPLEMENTATIONS, CLEARLY IDENTIFIED

DEPLOYED REFERENCE

Rules + BM25 k6 + one structured model call + release checks.

QUALITY CANDIDATE

Message-only route; retrieve if eligible; render approved response actions; audit the exact final reply.

HUMAN HANDOFF

Return a reason and a real support contact route. Never claim an action was performed.

ANNOTATION CHAIN

Evidence	Origin and review	Use
Historical 250	AI labels; repeatedly inspected	Training / regression
Frozen 200	AI labels; Arnav approved	Frozen benchmark
100 reply ratings	Astra xhigh; Arnav verified	Judge calibration audit
New 60	Astra xhigh before inference	One-shot confirmation

Sampling excludes prior candidate, golden, taxonomy and benchmark cases; confirmation also excludes development messages. Conversation overlap and $\geq 85\%$ fuzzy text matches are rejected. This measures distinct English complaints, not traffic-weighted demand.

The headline, with context.

The archived 200-case benchmark is kept immutable. New measurements do not overwrite it.

System	Accuracy	Macro-F1	Recall	Auto	Misses
Majority / escalate	11.5%	0.017	100%	0/200	0
Keyword / nearest	60.0%	0.601	34.6%	166/200	53
Same prompt, kO	80.0%	0.826	100%	0/200	0
Retrieval, k6	77.0%	0.789	93.8%	70/200	5

Read recall together with coverage. Always escalating misses nothing and resolves nothing automatically. The kO arm also has zero coverage because the citation gate vetoes uncited replies; it is not a clean retrieval-safety ablation.

0.789

Intent macro-F1

Paired bootstrap 95% interval: 0.647 to 0.836.

93.8%

Escalation recall

76 / 81 required escalations caught.
Bootstrap interval: 88.2% to 98.8%.

5 / 70

Misses among auto

A fluent automatic response can still require a human.

A HIGH JUDGE SCORE WAS NOT ENOUGH

The original Gemini judge scored agent replies 4.11/5 versus 2.69/5 for the simple baseline, but approved an obsolete limit and a fictional DM action. Preference order consistency was only 73.5%. Retrieval did not establish a classification improvement: k6 - kO accuracy was -3.0 points (95% interval -7.5 to +1.0).

Calibration audit: 100 exact replies on 50 matched messages were first rated by GPT-6 Astra at extra-high reasoning, then verified unchanged by Arnav Bule. Agreement against both Gemini orders: weighted kappa 0.137, exact 28%, within one 53%. The human saw AI scores; the stricter later rubric is a disclosed audit, not an independent blind replication.

Fix the reply, not the score.

Five recurring defects explain the candidate design. Known cases remain regression evidence.

01 Unsupported actions

h_168 claims a DM was sent; d2_003 claims feedback forwarding. Hypothesis: copied staff actions become promises.

Server-rendered actions prevent arbitrary operational promises. Rules and exact-text semantic review add checks.

02 Wrong-issue guidance

d2_007 confuses customer and Spotify-created playlists. Hypothesis: lexical overlap outranks issue fit.

Classify before retrieval. Review whether the exact proposed step addresses the customer's stated issue.

03 Stale procedures

h_003 repeats an old limit; d2_009/010 copy developer-work claims. Hypothesis: history becomes current policy.

Treat history as examples, not current policy. Dated official sources authorize a narrow set of present procedures.

04 Unhelpful handoffs

d2_010/d2_013 give no path forward. Hypothesis: abstention suppresses useful guidance.

Use an honest limitation and Spotify's verified support contact page; ask only necessary, specific questions.

05 Risk and ambiguity

h_079/h_105 expose frustration and churn misses. Hypothesis: coarse confidence masks escalation risk.

Keep rules, primary/secondary policy defaults and low-confidence escalation. A semantic checker cannot prove safety.

Review before release: the semantic check sees the exact finalized public text, including trimming and links. Failed checks produce a contact handoff. The candidate uses one to three calls; tokens and latency are measured rather than assumed.

Measure useful coverage.

Development selects a hypothesis. Fresh confirmation tests the frozen implementation.

30-CASE ROUTING EXPERIMENT / AI LABELS AND REVIEW

System	Recall	Auto	Useful auto	Tokens
Current	85.7%	26.7%	6.7%	77,127
Selective	100.0%	36.7%	3.3%	67,094

Selective routing missed 0/14 required escalations versus 2/14, but automatic decisions are not all useful replies. Its p95 was 3.35 s versus 3.17 s. The quality candidate adds approved actions and semantic review. Two weaker development iterations were rejected before the final design was frozen.

60 NEW CASES / FROZEN BEFORE INFERENCE / AI LABELS AND REVIEW

System	Recall	Auto	Useful auto	Misses
Reference	87.9%	36.7%	5.0%	4
Quality candidate	100.0%	6.7%	6.7%	0

Runtime	Reference	Quality candidate
p50 / p95	1.63 / 3.53 s	1.53 / 6.33 s
Input + output tokens	154,379	150,347
Model calls	60	104

Useful automatic coverage counts shipped automatic replies with grounding, safety and resolution/next-step scores ≥ 4 , no review flags, and response kind resolution or necessary clarification. All messages stay in the denominator. Safe handoffs are not counted as automatic resolutions.

Acceptance decision: Keep the deployed reference. Candidate promotion gates are not all met. New reply reviews are AI-authored and have not been verified by Arnav. The earlier human verification applies only to the original 100 ratings. Small samples and same-family generation/checking limit the conclusion.

Make the claim reproducible.

The evidence is worth more than the demo. Reproduce saved results without an API key.

WHAT IS MISLEADING ABOUT MY HEADLINE NUMBER?

Coverage hides utility. An automatic decision can be irrelevant, obsolete or unsafe. Report useful coverage and missed escalations alongside recall.

Confidence is not calibrated risk. A 0.90 intent threshold is a policy setting. The original selector used its F2 fallback, not the claimed recall target.

Labels and samples limit the claim. Original AI labels were reviewed by Arnav; new confirmation labels are AI-only. De-duplication and English sampling change the population.

Review independence matters. Arnav verified visible Astra ratings. The later rubric is stricter than the original judge. Agreement does not establish independent hand-labeling.

Replay is not a fresh model run. Saved receipts prove what ran. Inspected examples are regressions; timing covers successful local requests, not a production SLA.

WITH ONE MORE WEEK

Obtain an independent blind human pass on the new replies; calibrate routing against useful coverage; broaden current-source coverage with expiry checks; run a larger locked benchmark and a small deployment canary. Keep account actions and autonomous posting outside this prototype.

REPRODUCE / PYTHON 3.12 / ZERO NEW MODEL CALLS

```
python -m pip install -e ".[dev]"  
python analysis_tools/reproduce_benchmark.py  
python analysis_tools/reproduce_quality.py  
python -m pytest -q
```

README documents the under-15-minute replay path. The repository contains the full rubric, per-example rationales, failures, cached receipts, input hashes and 15 non-obvious decisions. The hosted app is a demonstration; form submission is intentionally left to the author.

[Evidence & decision log](#)

[Dataset: Customer Support on Twitter](#)

[Spotify: current playback guidance](#)

[Spotify: official support contact](#)